

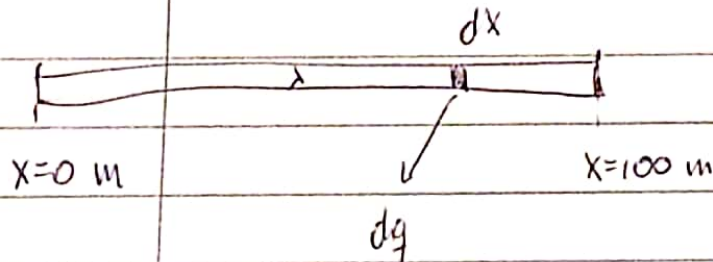
2.

$$\lambda(x) = 4\sqrt{x} \quad [\text{mC/m}]$$

$$\Rightarrow \frac{dq}{dx} = \lambda$$

$$dq = \lambda dx$$

$$Q = \int_0^{100} 4\sqrt{x} dx$$



$$= 4 \int_0^{100} x^{1/2} dx$$

$$= 4 \left. \frac{x^{3/2}}{3/2} \right|_0^{100}$$

$$= \frac{8}{3} (100)^{3/2}$$

$$= 2666,67 \quad [\text{mC}]$$

$$= 2,666 \quad [\text{C}]$$